DEBRE BERHAN UNIVERSITY

COLLAGE OF AGRICULTURAL AND NATURAL RESOURCE SCINCE

DEPARTMENT OF AGRICULTURAL ECONOMICS

THE ROLE OF MICROFINANCE INISTITUTION IN IMPRUVING HOUSEHOLD INCOME IN THE CASE OF DEBREBERHAN TOWN TEBASSIE SUB CITY & ATSEZEREYAEKOB ARESEARCH PAPER SUBMITTED TO DEPARTMENT OF AGRICULTURAL ECONOMICS PATRIAL FULFULMENT OF THE REQUIRMNET FOR BSc DEGREE IN AGRICULTURAL ECONOMICS.

GROUP MEMBERS

NO	NAME	ID	SEX
1	Wendmsew walelgn	DBU1501522	M
2	Petros Abiye	DBU1501423	M
3	Nigussie Demssie	DBU1501651	M
4	Mulu Engdaw	DBU1501383	M
5	Natnael Maru	DBU1501409	M
6	Rodas Hayilu	DBU1501436	M
7	Selam Girma	DBU1501757	F

Course Instructor: Mr. awgten

ADVISOR: Mr. Dawit

MAY, 2026

DEBRE BERH ETHIOPIA

ACKNOWLEDGEMENT

First of all, we would like to thank God for being with us in all aspects of our life especially in our academic performance. We would like to extend our heartfelt praise to our advisor Mr. Dawit. for his valuable advice, insight and guidance from the beginning to the completion of this research paper. Our praise also goes to department of agricultural economics for their assistance in the preparation of this research proposal through the costs needed in conducting the research. Finally, we would like to thank and appreciate the DBU department of agricultural economics and my advisor for their assistance in providing the important materials for us.

ABBREVIATIONS AND ACRONYMS

CSA	Central Statistical Agency
FGD	Focus Group Discussion
AMFI	Akufada Microfinance
IFAD	International Fund for Agricultural Development
MF	Microfinance
MFIs	Microfinance Institutions
Mo FED	Ministry of Finance and Economic Development
NGOs	Non-governmental Organizations
PPS	Probability Proportional to Size
DBU	Debre Berhan university

Table of content

Content page

ABBREVIATIONS AND ACRONYMS ..... **VI**
LIST OF TABLES ..... **VI**
LIST OF FIGURES ..... **VII**
ABSTRACT ..... **VIII**
INTRODUCTION **1**

1.1. Background **1**

1.2. Problem statement **2**

1.3. Objective of the Study **4**
1.3.1. General objective 4
1.3.2. Specific objectives 4

1.4. Research Questions **4**

1.5 Significance of the Study **4**

1.6. Scope and Limitation of the Study **4**
2. LITERATURE REVIEW **6**
2.1.1. Basic concepts and Definition 6
2.1.2. Sustainable livelihood framework 7

2.2. Empirical Review **9**
2.2.1. Determinates of household participate to microfinance Institution 9

2.2 The Contribution of Micro-Finance Institution in Building household income **10**
2.2.2. The role of microfinance on poverty reduction 10
2.2.3. Role of microfinance on livelihoods 11

2.3. Conceptual Framework **12**

3. METHODOLOGY 13

3.1. Description of the Study Area 13

3.2. Type of Research Design 13

3.3. Type and Sources of data 14

3.4 Sampling Techniques and sample size 14

3.5 Method of Data Collection 15

3.6. Methods of Data Analysis 16

3.6.1. Econometric Method of data analysis 17

3.7 Null Hypothesis 18

3.7.1 Dependent variable 18

3.6.2. Explanatory variables of the study 18

4. RESULT AND DISCUSSION 22

4.1. Demographic Characteristics of Sample Respondents 22

4.1.1. Sex sample respondents 22

4.1.2. Religion and Marital Status of Sample Households 23

4.1.3. Age, family size and age composition of sample households 24

4.1.4. Educational Status of Sample Household Heads 26

4.2. Major Sources of Household Income 27

4.3. Loan Purpose and Income Generating Activities of the Households 28

4.4. Household Income and Expenditure 29

4.5. Challenges of the Households in Credit Participation 30

4.5.1. Challenges related to transportation 31

4.5.2. Challenges related to the size of the loan and repayment time (due time) 31

4.6 Child Education and the Microfinance Services 32

4.6.1 Children Access to Education 32

4.6.2. Expenditure on Education 34

4.7. Logistic Regression **35**
5. CONCLUSION AND RECOMMENDATIONS **37**
5.1. Conclusion **37**
5.2. Recommendations **38**
6 REFERRENCES **39**
7 APPENDEX **40**

LIST OF TABLES

Table 2.1. Hypothetical relationship of variables……..11
Table 3.1 Sampling Techniques of the Respondents…….15
Table 4.1 Sex of the respondents……..23
Table 4.2. Religion and Marital Status of Sample Households……24
Table 4.3. Age, family size and age composition of sample households……..26
Table.4.4. Annual income, expenditure on food and non-food items……30
Table 4.5 Children at school age and children attending formal education……34

LIST OF FIGURES

Figure 1.1. Conceptual framework of the study 13
Figure 4.1 Source of income of the respondents 27
Figure.4.3, Loan Purpose of the Beneficiaries 22
Figure 4.4, Mean Annual Expenditure on Child Education. 36

ABSTRACT

Livelihoods are 'means of making a living', the various activities and resources that allow people to live. It comprises the capabilities, assets (including both material and social) and activities required for a means of living. Thus, microfinance programs have been considered as one of the main instruments in livelihood improvement in recent development agenda. Microfinance comprises the provision of financial services including credit and other facilities like savings, insurance, and transfer services to poor household. The study was conducted in Tebassie among target Branchs of the debreberhan . The main objective of the study is to identify the contribution of AMFI in improving household livelihood through providing credit. Further, the study used both primary and secondary data sources. The primary data was collected with semi-structured interview and focus group discussion from sampled respondents; whereas, secondary data gathered from different documents of the microfinance institution that exists in the study area. Stratified random sampling was used to select 60 respondents (35 beneficiaries and 25 non-beneficiaries) based on the principle of probability proportional to size (pps). Generally, Microfinance Institutions have an explicit potential that intends to improve the livelihoods of households. While improving livelihood, microfinance (AMFI) credit service increases the income of the clients; which, in turn, enable them educating their children and improves their nutritional status. Inferential statistics such as crosstabs; independent sample t-test and custom table are used in data analysis. Descriptive statistics like frequency, percentage, mean and standard deviation were employed to analyze the data. Chi-square (for categorical variables) and t-test (for continuous variables) were used to see the significance of the relationships of independent variables with dependent variables. Moreover, the qualitative data was analyzed by describing, summarizing and interpreting for further clarity. Finally, conclusion and recommendation were made based on the findings obtained from result and discussion.

Key words: ***assets, client, role, microfinance institution, livelihood,***

INTRODUCTION

1.1. Background

The origins of MFIs in Ethiopia is largely rooted in their NGO past with a clearly defined mission of rural poverty eradication. With a network of 500 sub-branches and branches, the MFIs have expanded their outreach to many of the regions where the incidence of poverty is highest. As of January 2001, MFIs has made loans to and mobilized savings from about 500,000 clients nationally. Some MFIs have also started to offer other services such as managing pension remittances and money transfer services At least 41% of the MFI clients nationally are women and in the majority from rural households. Most of the MFIs have two types of loan products, namely loans for on-farm activities, which are due in four to twelve months, and off-farm investments with more flexible repayments on weekly or monthly basis (IFAD, 2001). On average, 60% of the MFI portfolio represents loans for on-farm investments while income generating activities and petty trading accounted for about 40%. (Dejene, 2016).

Microfinance uses the likelihood approach to improve their livelihood by strengthening the following five basic assets and comprises broad and inter related programs and policies. These includes giving people salaried jobs and other opportunities to earn income; providing loans; saving and other financial services; providing training in jobs and business skills; developing institution alliance and network to advances economic interest, and promoting policy and social changes that improve the farming household livelihood Projects (Barry *et al*,2015).

According to a World Bank (2021), Ethiopia is one of the poorest and largest populated countries in Africa. Its total population was 84,734,262 in 2011; its economy is based on agriculture, which accounts for more than 35.45% of GDP, 80% of exports, and 80% of total employment. The biggest sources of foreign trade are coffee, flowers and oilseeds. Yet, in spite of high rates of growth most Ethiopians live in poverty.

The formal microfinance in Ethiopia started in 1994. In particular, the Licensing and Supervision of Microfinance Institution Proclamation of the government encouraged the spread of Microfinance Institutions(MFI) in both rural and urban areas as it authorized them among other things, to legally accept deposits from the general public (hence diversity sources of funds), to draw and accept drafts, and to manage funds for the micro financing business. Currently there are 29 licensed MFIs reaching about 905,000 credit clients and some saving clients. Considering the potential demand, particularly in rural areas, this only satisfies an insignificant proportion (Gobezie, 2021).

Debreberhan (AMFI) is currently operating largely in rural areas to complement the agriculture lead and rural centered development effect of the Federal Government of Ethiopia in general and Debreberhan in Particular. TEBASSIE District is also one of the places where Debreberhan Akufada microfinance inistitution is giving credit for the households.

1.2. Problem statement

In many developing nations, poverty and misery are common and most popular problem. As many studies revealed people living in less developed countries are under the line of poverty for many decades. According to World Bank report poverty has many dimensions, the first dimension of poverty is material deprivation caused by income and consumption shortages, the second dimension are: low achievement in education and health (World Bank, 2000). Third the low achievements in education, health and material deprivation made the society to be constant attached with situations of poverty, Ethiopia can be viewed as an important example of it .poverty in Ethiopia is manifestation of complex factors, such as high population growth, environmental degradation high unemployment, high level of illiteracy limited access to resource health and education services, etc. The poor in Ethiopia have low income that leads to low investment, which in turn leads to low productivity, and income. In previous time access to institution credit that contributes to an increase in investment is very limited in Ethiopia. The poor get access to financial service through informal channel such as money lenders, iddir, friends, relative, and traders with higher interest rate. But the financial services could not satisfy the potential demand of the poor for microcredit is enormous, because, every person needs income to activate. Ethiopia is predominantly an agricultural country that 85 percent of

its population depends on agriculture for their livelihood. Its agriculture is also characterized by low productivity, which can be attributed to shortage of skilled human, lack of good governance and lack of capital that compounded by inaccessibility of the poor to the existing formal financial institutions due to some factors such as high collateral (guarantee) requirements for small loans. Since poor people were regarded as credit worthless, access to credit is very limited (Gebrehiowot, 2012).

Relative to the experience of other developing countries, microfinance development in its institutionalized form is a recent phenomenon in Ethiopia. Lack of financial resources is one of the major problems facing poor households. Formal financial institutions are inefficient and inaccessible in providing credit facilities to the poor (Assefa *et al.,* 2015).

The core of this study is, therefore to investigate the impact of microfinance on improving household livelihood and to identify the major causes of microfinance North Shewa zone, Debreberhan town .These problem need serious attention in all organization which focuses for the improving household livelihood. So; the problem needs urgent response from development organization.

1.3. Objective of the Study

1.3.1. General objective

The purpose of the research is to Assess the role of micro finance institution on improving household income and to assess challenges of the users in credit participation in the study area.

1.3.2. Specific objectives

- To identify factor affecting of household participation to microfinance in study area.
- To assess the contribution of microfinance in increasing household income in the study area.

1.4. Research Questions

- What are the factor affecting of households in accessing microfinance in study area?
- What are the contributions of microfinance in increasing rural households income?

1.5 Significance of the Study

The product of the research identifies the role of AMFI in improving household livelihood in the study area. It also aims finding out the integration of the micro credit service and initiate other researchers to generate and add information on existing knowledge of micro finance contributions in improving the livelihood of households. Further, the findings of the research provide multipurpose information to different users, including practitioners in microfinance, donors, policy planners, academicians and the public at large. At worst, it raises questions for further research and study.

1.6. Scope and Limitation of the Study

The researches will be carried out on microfinance institutions (AMFI) that found in Tebassie Sub city. The study assessed and analyzed the role of AMFI toward improving the livelihood of household of the study area. And the extent to which it contributes in initiating and

improving the livelihood of household. The study includes the indicator values to identify the role of selected microfinance institutions on the improvement of household living standards in the study area. So, the scope of the study is at the household level.

The researchers were limited to deal with the role of microfinance institutions in improving household livelihood with specific microfinance institution; due to financial, time and labor constraints; that forced the researcher to limit the sample size and the scope of the research. So, the researchers focused only on the role of microfinance institutions in improving the livelihood of household (clients) of AMFI at tebassie branch. Generally the researchers faced problems like: lack of facilitated transportation, shortage of time and money, shortage of labor and lack of real information and etc.

2. LITERATURE REVIEW

2.1.1. Basic concepts and Definition

Microfinance as defined by (Asiama & Osei, 2007), encompasses the provision of financial services and the management of small amounts of money through a range of products and a system of intermediary functions that are targeted at low income clients through the provision of small loans and other facilities like savings, insurance, transfer services to poor low-income household and micro enterprises. Schreiner (2001) support this view by defining microfinance as "the attempt to improve access to small deposits and small loans for poor households neglected by banks."

Therefore, microfinance involves the provision of financial services such as savings, loans and insurance to poor people living in both urban and rural settings who are unable to obtain such services from the formal financial sector. Related concepts to microfinance are micro savings; micro insurance and MFIs. They are briefly explained in the subsequent paragraph.

Micro Savings is also a microfinance service that allows impoverished individuals to safeguard money and other valuables items and even earn interest. It allows a lump sum to be enjoyed in future in exchange for a series of savings made now (Arytery, 2008).

Micro Insurance is also a component of microfinance. It is the provision of insurance to households. Poor households are especially vulnerable to risk both in the form of natural calamities as well as more regular occurrences of illness and accidents. Microfinance Institutions (MFIs) have played an active role in reducing or protecting the poor households against this vulnerability. This is done by providing credit for increasing income earning opportunities and through providing savings services to build up resources that can be drawn down in cases of emergencies.

On the other hand, the terms micro credit and microfinance are often used interchangeably, it is important to highlight the difference between them because both terms are often confused. Sinha (1998), states that "micro credit refers to small loans, whereas microfinance is appropriate where NGOs and MFIs supplement the loans with other financial services such as

savings and insurance". Therefore, microcredit is a component of microfinance in that it involves the provision of credit to the poor, whilst microfinance add on non-credit financial services such as saving insurance, pensions and payment services (Ayertey, 2008).

Livelihood comprises the capabilities, assets (including both material and social res**o**urces) and activities required for a means of living. It can also define as the activities, assets and access that jointly determine the living gained by an individual or households. it is the means of support something that provides income to live on, a specially paid work to secure necessity of life through livelihood activities such as economic activities that people know own and undertake to earn income today and in the future.

A sustainable livelihood is that which can cope with and recover from stresses and shocks and maintain or enhance its capabilities and assets both now and in the future, while not undermining the natural resource base. Access to capital is one of the determinants of a household's ability to achieve well-being, defined broadly to include natural, physical, financial, human and social capital (Barry *et al.,* 1996).

2.1.2. Sustainable livelihood framework

Sustainable livelihood framework is a holistic, asset-based framework for understanding poverty and the working of rural poverty reduction. It also points out the need to building the assets of individuals and households. Moreover, this asset-access activities framework is iterative and is a process unfolding over time for individuals, households, and communities (Simon *et al.,* 200).

Asset Resources: Assets are resources owned, controlled, and claimed to access by the individual and the household who needed to cope with stresses and shocks, and to maintain and enhance capabilities now and in the future (Ellis, 2000). Resource assets can be natural, physical, financial, and social and human that contribute to a sustainable livelihood are interdependent (Lipton pers. Comm. cited in Scoones, 1998).

Natural capital comprises the land, water, and biological resources that are utilized by people to generate means of survival. Whereas *physical capital* is produced goods comprising building, irrigation canals, roads, power lines, water supplies, tools, machines, technologies and housing that are created by economic production processes. *Financial capital* refers to stock of money to which the household has access. It is chiefly likely to be cash, savings, and access to credit in the form of loans. Further, human capital is demographic such as gender structures, the body of education, skills, knowledge, ability to and good health and physical capability important for the successful pursuit of different livelihood strategies. So, human capital refers to the available to the household; its education, skills and health that is enhanced by investment in education, training and pursuing one or more occupations (Scoones, *et al.,* 2001). Social capital is the social resources (social networks, social claims, social relations, affiliations, associations more generally; and consensual norms and relationships of legitimate authority) upon which people draw when pursuing different livelihood strategies (Scoones, 1998: Ellis, 2000; Stephen and Simon, 2001).

2.2. Empirical Review

2.2.1. Determinates of household participate to microfinance Institution

According to Deaton 2013, The composition of household assets is found to be much more important as a determinant of household access to formal credit than the total value of household assets or landholding size. In particular, a higher share of land and livestock in the total value of household assets is negatively correlated with access to formal credit. However, land remains a significant determinant of access to informal credit. Therefore, poor households whose assets consist mostly of land and livestock but who want to diversify into nonfarm income generation activities may be constrained by lack of capital.

The results of logistic regression identify determinants of the rural households' participation in microfinance institution in the study area.. These include sex of household head, education level, cultivated land size, livestock holding, dependency ratio and frequency of extension contact. Accordingly, sex of household head had a positive effect on households' participation in microfinance institution and it was statistically significant at 10% significance level. The marginal effect of sex was 0.266. The value of marginal effect indicates that the probability of male-headed households' participation in microfinance services increases by 26.6% more than female-headed households, keeping other variables in the model constant. Moreover, the result presented the education level of the household head had a positive effect on the probability of participation in microfinance institution. But, the significant level was different with different levels of education. Here, from education category, illiterate was taken as the base category. Accordingly, there was a statistically significant difference between illiterate households and those households who have attended Primary education level at 10% significance level. The marginal effect of a primary education level was 0.254. The result of marginal effect implies that the probability of Those who have acquired a primary level households' participation in microfinance services increases by 25.4% than illiterate household. Similarly, the finding result indicates that there was a statistically significant difference between illiterate household and those households who have attended secondary education level regarding participation in microfinance services at 10% probability level. In

contrast, for occupation, the farmer was taken as a base category, but none of the category was significant. The marginal effect of sex was 0.266. The estimated marginal effect result shows that the probability of male-headed households' participation in microfinance services increases by 26.6% more than female-headed households, Keeping other variables in the model.

2.2 The Contribution of Micro-Finance Institution in Building household income

The MFI has a number of development assistance programs to promote community interaction and to sustain social contact either through extended family networks or geographic proximity as a means of building social capital through frequency of mandatory meetings with borrower groups. In addition, group lending, achieves clients to meet weekly in groups to make loan payments low rates of income, builds new social capital among participants and strengthening existing social capital. These meetings encourage regular interaction among members of highly localized communities. Encouraging clients to meet more often (for six to ten months), leads to persistent increases in social ties. Besides, social interactions encourage norms of reciprocity and trust. In fact, participation in groups is of tenused to measure an individual's or community's degree of economic cooperation (Feigenberget al., 2010).In addition, as a result of getting together, Isserles came with the data that those of contraception and child immunization had increased.

The contribution of MFIs funding to the rural and urban women and youth, business expansion is the vital role played by MFI. These institutions play a vital role in boosting up the incomes of their clients. At the MFIs services, the reason behind the massive use of the loan service as compared to the rest of the services is that most small scale entrepreneurs requirefunds for starting up ventures, reinvesting in the existing businesses or overcoming various financial obligations (Charles, 2003).

2.2.2. The role of microfinance on poverty reduction

Ditcher (1999), states that microfinance is a tool for poverty reduction and while arguing that the record of MFIs in microfinance is "generally well below expectation" he concedes that some positive impacts do take place. From a study of a number of MFIs he states that findings

show that consumption smoothing effects, signs of redistribution of wealth and influence within the household are the most common impact of MFI programs. Hulme and Mosley (1996) in a comprehensive study on the use of microfinance to combat poverty, argue that well-designed programs can improve the incomes of the poor and can move them out of poverty.

Wright (1999), states that there is significant difference between increasing income and reducing poverty. He argues that by increasing the income of the poor, MFIs are not necessarily reducing poverty. It depends on what the poor do with this money, oftentimes it is gambled away or spent on alcohol, so focusing solely on increasing incomes is not enough. The focus needs to be on helping the poor to "sustain a specified level of well-being" by offering them a variety of financial services tailored to their needs so that their net wealth and income security can be improved.

2.2.3. Role of microfinance on livelihoods

Having access to MF services have led to an enhancement in the quality of life of clients, had increased their self-confidence, and had helped them diversify their livelihood security strategies and thereby increase their income (Robinson, 2001). Health care and education are two key areas of non-financial impact of MF at a household level. Wright (2000) stated that from the little research that has been conducted on the impact of MF interventions on health and education, nutritional indicators seem to improve where MFIs have been working. MF interventions have been shown to have a positive impact on the education of clients‟ children because one of the first things that poor people do with new income from micro-enterprise activities are to invest in their children's education (Littlefield *et al.,* 2003).

Moreover, women empowerment is also a key objective of MF interventions. Women need empowerment as they are constrained by the norms, beliefs, customs and values through which societies differentiate between women and men. MFI cannot empower women directly but can help them through training and awareness rising to challenge the existing norms, cultures and values that place them at a disadvantage in relation to men and to help them have greater control over resources and their lives as *Kabeer*, quoted in (Mosedale, 2003).

2.3. Conceptual Framework

Conceptual frame work is the maps that showing the interaction of the concepts. It would mobilize the habit of microfinance service. Creating better way better for achieving means of living for the beneficiary or clients of AMFI that found in the rural through institutional credit, saving, insurance and generating employment opportunities, education opportunities, nutrition facilities. Diagrammatically, it can be shown as follow:

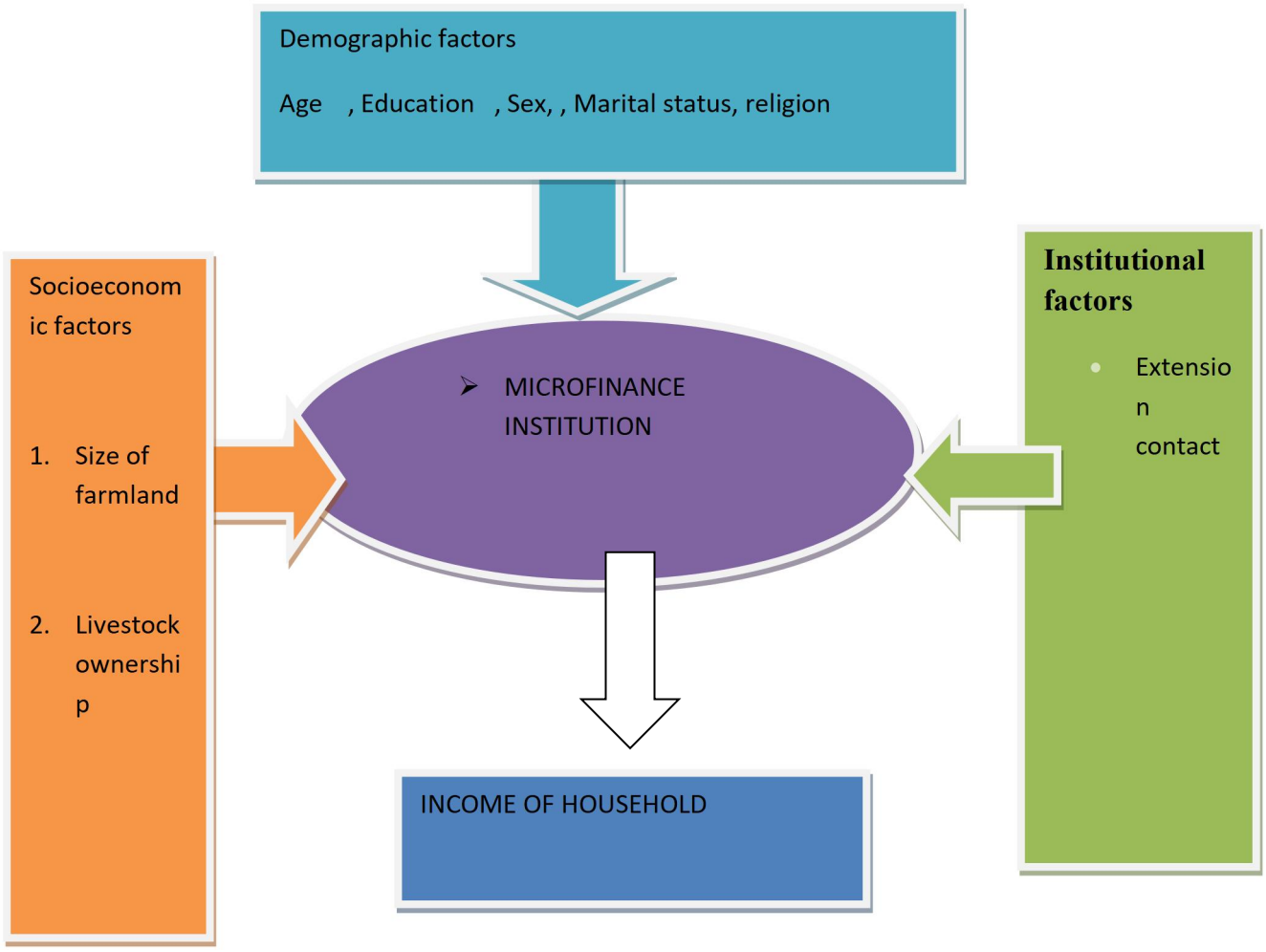


Figure 1 conceptual framework

3. METHODOLOGY

3.1. Description of the Study Area

The study was conducted in Debreberhan town. Debreberhan is one of the towns in the North Shewa Zone, Amhara Region of Ethiopia. This town, which is located 130 km northeast of Addis Ababa, is bordered in the north by the Basona Warana district Saria Branch, in the south by the Basona Warana district Kor Margefinaya Branch, in the east by the Basona Warana district Wushawushign Branch, and in the west by the Basona Warana district Angolela Branch. Also, this city is divided into 5 sub-cities (Atse Zereyacob, Etege Tayitu, Emeye Minilik, Tebassie, and Chacha sub-city), 9 urban Branchs, and 5 rural sub-Branchs. The 2007 census conducted by the CSA of Ethiopia, updated in 2021, estimates a total population of this city of 188,513, of whom 89,579 were men, 98,934 were women, and 8,906 were households, situated in a total area of 18,081.95 hectares.

Debreberhan town is located at 9° 40' N latitude and 39° 30' E longitude with an elevation of 2840 meters above sea level. The terrain is 86% plain, 1% mountainous, and 4%, and the soil type is bitter, red, and brown. The average annual temperature of the city during the day and night is 20.7°C and 8.2°C, respectively, with precipitation of 964 mm. (Debreberhan Town Administration Office, 2021)Land used for agriculture by the city administration is 3485 hectares, and total forest cover is 29.955 hectares. According to Berhan, in terms of manufacturing, more than 500 hectares of land have been arranged in four clusters. These are the agro-processing, wood and metal, textile, garment, chemical, and construction sectors.

3.2. Type of Research Design

A cross-sectional type of research design will be used, which employed comparative study between participants and non- participants of micro finance scheme. Furthermore, descriptive survey type of research will be used. Cross-sectional survey design was selected to collect data from the sample population at specific point in time and based on the results to make generalizations. Descriptive survey research will be chosen to generally describe the differences experienced by households aroused from obtaining microfinance services in comparison with non-beneficiary households.

3.3. Type and Sources of data

Two types of data sources, primary and secondary will be used to obtain the desired qualitative and quantitative data types in order to meet the study purposes. Primary data has been obtained from the potential informants (beneficiaries and non-beneficiaries) and the micro finance specialists who were working for the organization delivering the scheme. The secondary data are further gathered from document of the microfinance institution that exists in the study area.

3.4 Sampling Techniques and sample size

The sample frame comprised of two categories: the microfinance participant and non-participant households. The sampling frame was compiled by drawing together lists of names of beneficiaries obtained from the microfinance documents who were providing micro finance service to them. For non-participants names were obtained from village savings and credit cooperatives using credit seekers data.

The total number of target population is 580 i.e. the number of households registered in credit seeker data and the farmers of the tebassie in the two Branches. From the total of.580 target population 85 respondents were selected randomly as a sample. Hence the corresponding number of respondents from the two Branch was obtained based on the principle of probability proportional to size (pps) as it is indicated in the table below.

	Beneficiaries		Non-Beneficiaries		
Branch	No of clients	Sample taken	No of credit seeker	Sample taken	Total
Tebassie	162	28	118	26	54
Atsezereyaekob	180	18	120	13	31
Total	342	46	238	39	85

Table 3.1 Sampling Techniques of the Respondents

Regarding number of sample size the researcher decides to take the following formula.

We use Yamane's formula

Sample Size=$\frac{\text{Target population}}{1+ \text{target population (e) } 2}$

$$n = \frac{N}{1 + N(e^2)}$$

Where: n = statistically acceptable sample size

N = Total size of target population

e = level of precision (depends on level of significance). (10%)

The participants in the micro finance service will be select from the lists using stratified random sampling method, based on the loan cycle of the beneficiaries. AMFI provides microfinance credit and other services for 3 Branchs. Among these Branchs, two Branchs tebassie and atsezeryaqob were selected through random sampling techniques. Purposive stratified random sampling techniques were used to select a total of 60 households (46 households participating in micro finance scheme and 39 non-participants) for the study. Those participants who have reached second the loan cycle and above have been targeted to study the role of the tebassie scheme in improving rural household livelihood.

3.5 Method of Data Collection

Primary data will be gathered through informal and formal survey. Informal survey was under taken first; to collect background information which was useful for subsequent survey. Then formal survey was conducted to assess the role of microfinance institution in improving rural household livelihood in the study area by using open ended and closed ended questions for semi-structured interview schedule and checklist for focus group discussion. The interviews were help to gather the necessary qualitative and quantitative information through asking questions and writing down the response of the respondents which build research purpose. It was proposed to those people select as a sample. On the other hand, focus group discussion was used by the researcher to obtain qualitative data. FGD allowed a dialogue among participants and stimulates them to openly express their views on the issues raise

Secondary data will be gathering through reviewing of documents, reports and records of published and unpublished documents. It is the main source of information and these data were easily available inexpensive and obtain quickly. These secondary data indicate the past and current performance of microfinance institutions in improving livelihood through providing microfinance credit service.

3.6. Methods of Data Analysis

The quantitative data, which will be collected, based on semi-structured interview analyzed by using inferential statistics like cross tab, Custom table and independent sample t-test. Descriptive statistics such as frequency, percentage, mean and standard deviations were used to analyze the quantitative data. Chi-square (for categorical variables) and t-test (for continuous variables) were used to identify the significance of the relationship of independent variables with dependent variables. It includes the comparison of demographic factors, income, sources of income, and household income and expenditure on food items & non-food items using SPSSS. Moreover, the qualitative data was analyzed by describing, summarizing and interpreting for further clarity.

3.6.1. Econometric Method of data analysis

The modeling methodology used to analyze the factors determining household's participation in microfinance program is logistic regression that allows the prediction of discrete variables by a mix of continuous and discrete predictors (McCullagh and Nelder, 1983). The model constrains the estimated probabilities to be either 0 or 1, relaxes the constraints that the effect of independent variables is constant across different predicted values of the dependent variable.

The logit model assumes that while only the values of 0 and 1 are observed for the variable Y , there is a latent, unobserved continuous variable Y* that determines the value of Y. It is assumed that ε_i is normally distributed across observations, and the mean and variance of ε are normalized to 0 and 1. The Y* can be specified as follows:

$$Y^*_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots + \beta_k x_{ki} + \varepsilon_i \quad \ldots\ldots(1)$$

And that: $Y_i = 1$ if $Y^*_i > 0$ $Y_i = 0$ otherwise. It is assumed that y_i^* is a function of observed and unobserved variables

$$y_i^* = \beta_0 + x_{1i}\beta_1 + x_{2i}\beta_2 \ldots. + x_{ki}\beta_k + \varepsilon_i \quad \ldots\ldots(2)$$

$$y_i^* = x_i\beta + \varepsilon_i \quad \ldots\ldots(3)$$

Where: y_i^*= latent and measure of credit market participation by *i th household*,

Xi = a vector of explanatory variables describing the personal, social, economic and environmental factors,

β_i= a vector of parameters to be estimated, and

ε_i = a random error term (assumed to follow a standard normal distribution).

The model is determined by the assumed distribution of ε. The observed and coded discrete credit market (MFI) participation variable, y_i^*, is determined from the model as follows:

$$\Pr(y_i=1) = \Pr(\varepsilon i \geq -(\beta_0 + x_{1i}\beta_1 + x_{2i}\beta_2 \ldots. + x_{ki}\beta_k))$$

$$= 1 - \Pr(\varepsilon i \leq -(\beta_0 + x_{1i}\beta_1 + x_{2i}\beta_2 \ldots. + x_{ki}\beta_k))$$

$$= 1 - \Phi(-x_i\beta)$$

Where, Φ represents the cumulative normal distribution function. The interpretation of this model's primary parameter set, β, is as follows: positive signs indicate likely factor for household's participation in credit program as the value of the associated variables increase, while negative signs suggest the converse.

The model considers households as utility maximizes who compare and rank level of utility that they get by household's participation in credit program against its reservation utility attained without participation. In general, the effect of a change in one of the explanatory variables (say the *j*-th variable) on the choice probability is given by the derivative.

3.7 Null Hypothesis

3.7.1 Dependent variable

The dependent variables for the logit/probity analysis have dichotomous nature representing rural household participation in micro financial institution (Household income). This is to distinguish or discriminate between those participants and non-participants in the study area. It takes value of "1" for participants "0" for non-participants in micro financial institution

3.6.2. Explanatory variables of the study

To establish working hypotheses of this study different literatures on factors influencing household participation in microfinance institution, past research findings and the author's knowledge of microfinance institutions in the study area are used.

1. Age of the farm household head (AGE): It is a continuous variable, defined as the farm household heads age at the time of interview measured in years. According to (Dejene Aredo, 1999) those farmers having a higher age have better knowledge to the importance of micro financial institution due to life experience and they have better association with micro financial institutions. Dercon et al. (20019) also says that As age of household heads increases households' to participate in micro financial institution also increase.

2. Gender: This is a dummy variable that assumes a value of "1" if the head of the household is male and "0" otherwise. Since women's lack of control over economic resources and the

nature of their economic activity and also male headed households have mobility, participate in different meetings and have more exposure to information.

3 Marital status: It is a categorical variable represented by 1 if the member is married, 0,2,3 otherwise. *According to Arega Bazezew* **&** *Wubliker Chanie (2015)* married household heads were better participated in *microfinance* than unmarried household heads. *On the other hand* Chambo, 2009 and Rahman, 2010) assumed that married households can handle many economic activities and expend more than households who are divorced, widowed, or single. Then the expected relationship between marital status and participation in microfinance institution will be either positive or negative.

4. Religion is a dummy variable which takes a value "1" and "0" for Christians and Muslim respectively. Microfinance institutions have highly relationship with all religions. Therefore, it is expected to that all religions have great efforts to affect microfinance institutions.

5 Education level (EDUCATION): Refers to the number of years of formal schooling the farmer attended. The higher the education level, the better would be the knowledge of the farmer towards the microfinance and acquire news and education about the benefits of the microfinance easily (Kraenzle,2017; Klien *et al.,2018*). So this variable is expected to influence the income positively

6 Family sizes: This variable is a continuous explanatory variable and refers to the total number of family in the household. When family size increases, their income also increases. This could be due to the number workers increased in the business activities so that it adds value in to their income increment (demise, 2016). So we expect family sizes with income and have positive relationship.

7 Farm size (FARMSIZE): This variable is a continuous variable and it refers to the total area of farmland that a farmer owns in hectare. It is assumed that the larger the total area of the farmland the farmer owns, the higher would be the output. Farmers with higher level of output expected to have higher income than those who have not. Therefore, it is expected that this variable would have positive influence on the income of cooperative members.

8 Total livestock holding (TLSH): This variable is a continuous variable and refers to the total number of livestock that the household own. It is assumed that household with larger livestock have better economic strength and financial position (Techane, 2002; Teferi, 2003 Therefore, this variable has positive association with income.

9 Number of years of membership (MEMBERSHIP): This variable is a continuous variable and it refers to number of years since the farmer has been the member of the microfinance. Farmers having longer years of membership are in a better position to know the benefits of the microfinance than farmers with shorter years of membership (Cain *et al.,2008*). In this study, this variable is hypothesized to influence income positively.

1O Additional benefit (PATREF): It is a continuous explanatory variable and refers to the amount of money the farmer receives from the surplus the microfinance appropriate in proportion with his/her use of the microfinance. It is assumed that farmers will be encouraged to produce and market more of his products through the microfinance if there is surplus appropriation in the form of microfinance refund (Black and Knutson,2007; Fulton and Adamowicz,2009). Thus, microfinance refund assumed to influence the income positively.

11. Size of land: It is the total cultivated land holding by the household in hectares. It is a continuous variable. The larger the cultivated land size the more input used that demands additional capital that might be obtained through credit from the available informal financial institutions (Sissay, 2008; Lensink*et al*., 2009). According to Babatund *et al,* 2007 and Bime&Mbanasor, 2011 a farmer with a large farm size has a better chance to earn more income which in turn enables them to save more .Therefore, it is hypothesized that larger size of land will affect participation in informal financial institution positively.

There are independent (explanatory) and dependent variables in the hypothesis

Table 2.1. Hypothetical relationship of variables

Independent Variables	Effect of independent variables on dependent variables
Age	Positive
Marital status	Positive /Negative
Educational status	Positive
Religion	Negative
Dependency ratio	Negative
Asset ownership	Positive
Gender	Negative
Membership	Positive
Livestock holding	Positive
Farm size	Positive
Family size	Positive

4. RESULT AND DISCUSSION

4.1. Demographic Characteristics of Sample Respondents

4.1.1. Sex sample respondents

In order to analyze the role of MFI AMFI) on the livelihood, the sampled respondents were classified into beneficiaries and non-beneficiaries. They are further categorized according to their Branchs due to there is some difference among respondents in terms of sex in the two Branchs.

Table 4.1 Sex of the respondents

Description		Tebassie		Atsezereyaekob			
		Beneficiary	Non-Beneficiary	Chi-square	Beneficiary	Non-Beneficiary	
		Column N %	Column N %		Column N %	Column N %	X^2-square
Sex of the Respondent	Male	40.0%	13.3%	2.727*	13.3%	26.7%	0.833
	Female	60.0%	86.7%		86.7%	73.3%	

*, *statistically significant at 10%*

Source: Primary Data

In the table 4.1, the proportion of female respondents in Tebassie Branch is 60 percent as compared to male respondents (40%). Similarly, the proportion of male respondent of non-beneficiaries is much less than the proportion of the female respondents in this Branch. Moreover, for both the beneficiary and non- beneficiary respondents in the Atsezerayaqob Branch, the male respondents' proportion is also lower than female respondents. From this data 76.7% of the respondents are female. This implies that MFI are working for the empowerment of women, the poor group of the society, in economical, social and on other aspects. Similarly, evidence from Mayoux (1997) indicates access to credit was seen as vital to women's ability to earn income, to wider their status and autonomy.

4.1.2. Religion and Marital Status of Sample Households

Religion and Marital status of sample household are the aspects of determining demographic characteristics of sample households. Review of related literatures emphasis, religion and marital conditions of households influence the income and the consequent livelihood situation of the family. In this research, religion and marital statuses of sample households were assessed and the result is displayed in the table 4.3 below.

Table 4.2. Religion and Marital Status of Sample Households

Description	Tebassie			Atsezereyaekob		
	Beneficiary Column N %	Non-Beneficiary Column N %	Chi-square	Beneficiary Column N %	Non-Beneficiary Column N %	Chi-square
Marital Status						
Married	86.66%	60%	2.861	86.67%	73.33%	6.167**
Divorced	6.67%	13.33%		0	26.67%	
Widowed	6.67%	26.67%		13.33%	0	
Religion						
Muslim	66.67%	100%	6.00**	80%	80%	0.667
Orthodox	26.67%	0		13.33%	6.67%	
Protestant	6.67%	0		6.67%	13.33%	

Source: Primary Data **, *statistically significant at 5%*

Table 4.2 reveals, although the microfinance beneficiaries are the women groups, 86.66% of them for TEBASSIE and 86.67% for ATSEZEREYAEKOB Branchs are married. This indicates married respondents have good social relationship within the society; as a result they get loyalty from the MFI scheme and given credit. Similarly, married non-beneficiaries in the two Branchs are 60% (TEBASSIE) and 73.33% (ATSEZEREYAEKOB). Since the sample of none beneficiary respondents is taken from the village credit seeker data, this indicates the

social acceptance of married households in one aspect and on the other hand this could denotes the married households need credit to meet the needs of their family. The percentage of divorced beneficiaries in the two Branch's is lower as compared to the percentages of non-beneficiaries of the two Branch's. This might be because of those divorced households are considered as poor in social relationship by the society. Hence, they fear to include in their group. Similarly, the percentage of widowed beneficiaries in the two Branch's is lower relative to the percentages of the non-beneficiaries in the two Branchs. This implies widowed households are less likely to participate in credit program; this might be due to fear associated with the repayment of the loan.

The result of the chi-square indicates that there is significant relationship between beneficiaries and non-beneficiaries in terms of their marital status. The result indicates that married respondents have greater tendency to participate in the credit program than divorced and widowed respondents. This might be related with preference of microfinances that married clients are more responsible in managing the finance and this may reduce the default rate for the microfinance.

On the other hand, around 70 percent of beneficiaries are Muslims; whereas around 90% of non-beneficiaries are Muslim. The percentage of non-beneficiary Muslims is higher than the percentages of beneficiaries. From the whole respondents around 81.7% are Muslim. The result of the chi-square also indicates as the number of Muslim respondents' increases, the chance to participate in the credit program decreases. But, the percentages of Orthodox and protestant are higher in beneficiaries than non-beneficiaries. This may be due to religious view towards the system.

4.1.3. Age, family size and age composition of sample households

Family size and age composition of households determine household livelihood. When there are extended family members and broad based age distribution, there is a need to get credit in order to meet the needs of the family i.e. food, shelter, clothes, health and primary education. The extended family members require more expenditure for food and less for savings and

investment. The following table illustrates the mean family size and age composition of respondents for both beneficiaries and non-beneficiaries.

Table 4.3. Age, family size and age composition of sample households

Description	Beneficiary		Non-beneficiary		
	Mean	St.deviation	Mean	St.deviation	t-value
Age of the respondent	37.83	9.731	38.80	11.146	-0.358
Family size	6.3333	2.02286	6.3333	2.07337	0.000
age group under15	3.1000	1.97135	3.2333	1.63335	0.143
age group 15-65	3.2333	1.54659	3.2333	1.63335	0.000
Olds	0.0000	0.00000	-0.0333	0.18257	-1.000**

Source: Primary Data ***, statistically significant at 5%*

From the table 4.3, both t-value and p-value indicates even though there is difference between beneficiaries and no-beneficiaries in terms of their age of the respondents, the variation are not significant.

On the other hand, the result of the t-test implies the mean difference between beneficiaries and non-beneficiaries at old age is -0.03333. And it is statistically significant at 5%. This shows at old age, when the household increase by one unit of age, the likelihood of credit participation decreases by 0.3333.

4.1.4. Educational Status of Sample Household Heads

In most developing countries illiteracy rate is higher especially in rural areas Shimelles and Zahidul, (2009). Since Ethiopia is developing country, illiteracy is common among the rural societies. Similarly, from the field survey, illiteracy rate is higher as compared the other educational levels in the Tebassie and Atsezereyaekob Branchs' respondents.

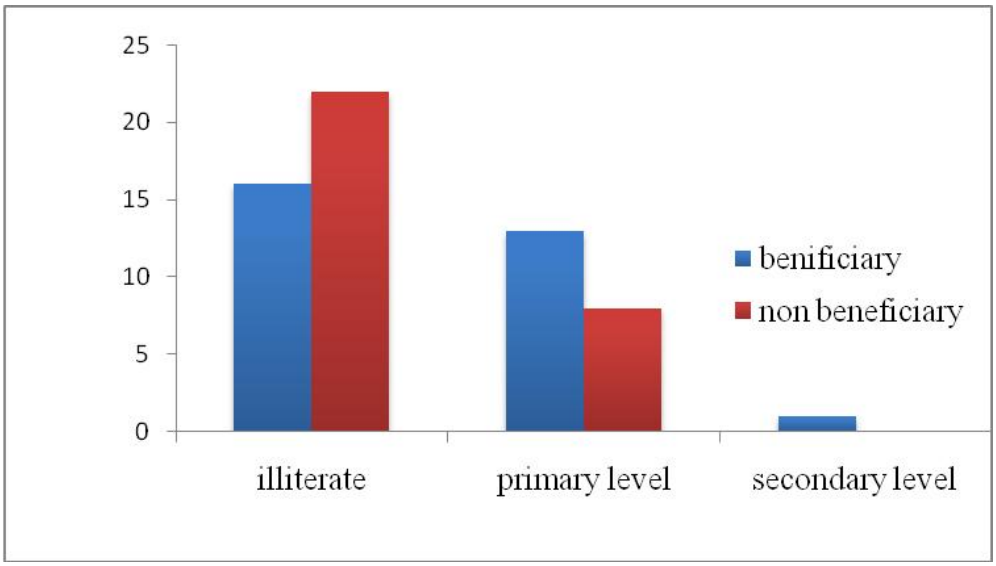


Figure 4.1 Educational levels of the respondents.
Source: Primary Data

As indicated in the figure above, from the total respondents around 63.3% are illiterate and around 35.0% and 1.7% are in primary and secondary level respectively. Comparisons were made between beneficiaries and non-beneficiaries of microfinance schemes in Tebassie and Atsezereyaekob Branchs, the number of illiterates are highly seen in the non-beneficiary part of the respondents. This is to mean that beneficiaries in the two branches have low illiteracy rate relative to non-beneficiaries. Moreover there are higher number of respondents with primary and secondary school enrollment in the beneficiary's category as compared to non-

beneficiary respondents. From the result of chi-square (p-value of 8.4%) there is significance relationship between educational status and credit participation. The result implies, literate respondents have high tendency to participate in the credit program than illiterate respondents. This might be MFI scheme considers educated households are economically active; and they are assumed better in diversifying their income sources through engaging on different income generating activities than the counterparts.

4.2. Major Sources of Household Income

The household income is determined by the sources of income. In this regard, attempt has been made to identify the major sources of incomes of the sample respondents in terms of beneficiary and non-beneficiary. The major sources of household income are depicted in the chart below.

Figure 4.2 Source of income of the respondents

Source: Primary Data

The above figure illustrates the major source of income for both beneficiaries and non-beneficiaries is farming. However, when comparison was made between beneficiaries and non-beneficiaries in terms of income sources, beneficiaries engaged on small-scale trade in addition to farming. This indicates beneficiaries run both farm and non-farm activities by using the credit they received from AMFI. So, their sources of income are farm income and non-farm income. On the other hand, about 95% of non-beneficiaries depend on farming for their income. The remaining 5% of them get income from labor and remittance. From this even though, the extent is very small non-beneficiaries get income from different sources as compared to beneficiaries. This indicates households who get income from different sources are not vulnerable for different problems associated with income constraints. Therefore the probability of their participation in MF credit program for diversifying their income decreases. From the result of chi-square (10.748^{a}), at 5% significance level, there is variation between beneficiaries and non-beneficiaries in terms of source of income.

4.3. Loan Purpose and Income Generating Activities of the Households

Beneficiaries of the MFI scheme AMFI) borrow money for different purposes. The Figure below shows the loan purpose of the beneficiaries.

Figure.4.3, Loan Purpose of the Beneficiaries

Source: Primary Data

As it is clearly indicated in the above figure , beneficiaries borrow money for input purchasing, livestock fattening and running small trading activities. The figure shows beneficiaries use 45% of the loan for purchasing input which used in agricultural activities; 32% ad 23% for livestock fattening respectively. This shows beneficiaries use greater the loan for input purchasing. From the survey the loan given by the MFI is small to run business which needs high investment cost. As the strategy of mitigating the problem beneficiaries engaged in low initial cost business activities. The result of the figure above shows this fact too.

From the Fig.4.2, the major source of income for both beneficiaries and non- beneficiaries is farming. The income generating activities, in which non- beneficiaries engaged are mainly crop production such as; -, cereal production (maize and teff), carrot, fruits and vegetable production. On the other hand, beneficiaries have been engaged on small-scale trade in addition to farming. From farming they produce, maize, teff, carrot, fruits and vegetables like tomatoes and potatoes. They also have been engaged in livestock fattening by using the loan received from the AMFI. Moreover, apart from farming, they also engaged in non-farm activities like small shopping and purchasing grains during the harvesting time and sell during the cropping season.

Generally, 80% of beneficiaries are participating on farming as income generating activities for their business. Evidence from Dejene (1999) stated this fact on his research “the Village Economy and Indigenous Financial Institutions”; ‘on average, 60% of the MFI portfolio represents loans for on-farm investments while income petty trading accounted for about 40%’. This is might be due to the loan given by the institution is small to run non-farm business activities, which needs high investment, cost as compared to agricultural business type. Secondly, it may be due to the site in which they are living is rural area.

4.4. Household Income and Expenditure

In addition to the household sources of income and income generating activities, the amount of income and expenditure determine the household livelihood. Based on these, data related to household income and expenditure was collected from both beneficiaries and non-beneficiaries for the purpose of comparisons. Major household expenditures were classified into food and non- food expenditures. Therefore, table 4.4 shows the mean monthly income, expenditure on food and non-food items.

Table.4.3. Annual income, expenditure on food and non-food items.

Description		Mean	Std. Deviation	T-value
Total income in 2016 fiscal year	Beneficiary	5.0333E3	1828.57271	4.136**
	Non-beneficiary	1.0047E4	6382.34274	
Expenditure on food	Beneficiary	2.0933E3	692.28922	1.375
	Non-beneficiary	1.8383E3	743.78845	
Annual cost of labor	Beneficiary	250.00	661.633	0.042
	Non-beneficiary	243.33	571.558	
annual cost of inputs	Beneficiary	1076.67	614.022	1.366
	Non-beneficiary	880.00	494.382	
Annual cost for medical services	Beneficiary	676.67	476.288	0.552
	Non-beneficiary	608.33	483.358	
Annual cost for education	Beneficiary	738.33	362.150	2.110*
	Non-beneficiary	650.83	503.200	

Source: Primary Data **, *statistically significant at 5%*

From the above table, the mean annual income of beneficiaries' in 2016 fiscal year is 5.0333E3; whereas the mean annual income of the non-beneficiaries in 2016 fiscal year is 1.0047E4. From the above illustration in the form of table, beneficiaries earned higher income as compared to non-beneficiaries in this specified fiscal year. This might indicates the role of credit received from the AMFI in increasing the income of the households. This in turn improves the livelihood of the households. The t-value clearly shows there is significant difference between beneficiaries and non-beneficiaries in terms of their income at 5% level of significance. Evidence from (Robinson, 2001) also stated that having access to MF services have led to an enhancement in the quality of life of clients, had increased their self-confidence, and had helped them diversify their livelihood security strategies and thereby increase their income.

On the other hand, there is expenditure on food items as well as expenditure on non-food items such as expenditure for purchase of input, medical services; for education and labor (wage). As indicated in above table, beneficiaries expend on food and non-food items more than non-beneficiaries. This might be because of beneficiaries have higher income as compared to non-beneficiaries. As per the survey, the nutritional trends of the beneficiaries improved by consuming different varieties of food due to improvements in business profit and high income from the sales of coffee and chat.

4.5. Challenges of the Households in Credit Participation

The main reason to classify the respondents into beneficiaries and non-beneficiaries is their tendency of participation in credit program delivered by the AMFI microfinance. So, the respondents are grouped into beneficiaries and non-beneficiaries. As per our survey, non-beneficiaries are not taking part in the cedi participation due to the following reasons.

One; Access to loan from individual lenders (relatives and neighbors)

Two; Fear to repay the loan back (due to risk associated with production activities)

Three: Information gap (lack of awareness the purpose and the aim of AMFI)

Four: The size of loan given by the AMFI is small to engage in business activities.

Even though all of the above factors are cited as a reason for not participating, most of them put information gap as the main reason for not participating.

On the other hand, clients are participating in the credit program because of: the interest of AMFI is better as compared to the interest of the individual lenders; the amount of loan given by AMFI is higher as compared the amount given by the individual lenders; in addition to credit provision, AMFI is giving training how use the loan and encourages for saving.

4.5.1. Challenges related to transportation

According to the survey, AMFI is found in the town while most of the clients are farming community in the rural areas. This is a challenge for these farmers to reach on time to get the service in the town. As they said and as the researchers seen practically, there is no a facilitated road which link them with the town in which AMFI is operating. In spite of this, if the clients arrive late they have been punished for being late. This is a double challenge for them; they travel long journey on their foot; subsequently, they are punished economically for being late.

4.5.2. Challenges related to the size of the loan and repayment time (due time)

As the researcher collected secondary information from the document of the organization, the minim size of the loan is 2000 birr while the maximum loan size is 10,000 birr. Clients are expected to pay interest of 150 birr per year for each 1000 birr they lent. The loan size depends on, (I) the period in which the clients stay by participating in the credit program (i.e. the client, who participated for longer years will get higher loan as compared to the beginner clients). (II) In addition, the loan size is determined by the past performances of the clients. Before taking the loan, each client proposes his/her plan. Depending on their proposal the loan is given to them; then at the end of the year, the staff members crosscheck their performance with what they had planned to do. If their performance is good the staff members encourage them to do more and the organization provides higher loan than what they received before. On the other hand, if their performance is not good the staff members decided to give them training to enhance their future performance.

However, there are respondents who complained the size of the loan given by the AMFI is not enough to carry out what they have planned to do. Moreover, there are also problems related to the repayment period. Most of them are engaging in agriculture. Among the factors affecting agricultural production: lack of adequate technology and product instability (yield variation) resulted from seasonal weather to variation are the common. If they get higher profit in one year, they will get low profit for the next year. Similarly if the weather condition is good they produce higher out and vice versa. Evidence from "the Contribution of Microfinance Institutions to the Economic Activities of the Youth and Women in Luwero District in Uganda" conducted by (Charles, 2003) also put high interest rates, short repayment periods, and lack of collateral security and inaccessibility of microfinance services as problems encountered in using microfinance credit service.

The other farm business in which they have been engaged is livestock fattening. This requires longer time period to get higher profit, but the due time given by the AMFI is only one year period. In addition there is the probability of the livestock to die, which is a risk for the clients. Furthermore, due to they provide group responsibility as guarantee to take the loan, if one of the group member cannot pay back the loan, they are forced to pay temporarily; then the MF scheme reports to the local leader in which the individual (one who refused to pay) is living. Then local leader forces that individual to repay the loan back. The reason that the AMFI reports to the local leader is, when the individuals come to the microfinance to be client, first they brought letter which ensures his/ her status. Even though this is the agreement that they have with the AMFI, it is one of the challenging conditions to the participants of the program

4.6 Child Education and the Microfinance Services

4.6.1 Children Access to Education

Access to microfinance schemes has an impact on the educational attainment. This means that, the income improved through access to the service enable the poor households educate their children. Even though the type of education and job availability matters, it is believed that the educated people have wider opportunities for access to skilled and semi-skilled jobs.

Therefore, they can generate better incomes, which improve their purchasing power and the subsequent livelihood improvement. Furthermore, these people are largely absorbed in the different sectors of the economy. Based on these assumptions, an attempt has been made to investigate whether the provision of microfinance services make differences on the child education between beneficiaries and non-beneficiaries in the study areas. Hence, the following table portrays the number of school age children (5-17 years) and those currently attending formal school respectively.

Table 4.4 Children at school age and children attending formal education

Description	Beneficiaries		Non-beneficiaries		
	Mean	St.deviation	Mean	St.deviation	t-value
Children at school age	3.40	1.070	3.97	1.712	-1.538*
Children attending formal school	2.47	0.860	1.86	0.915	2.615

Source: Primary Data *, *statistically significant at 6.2%*

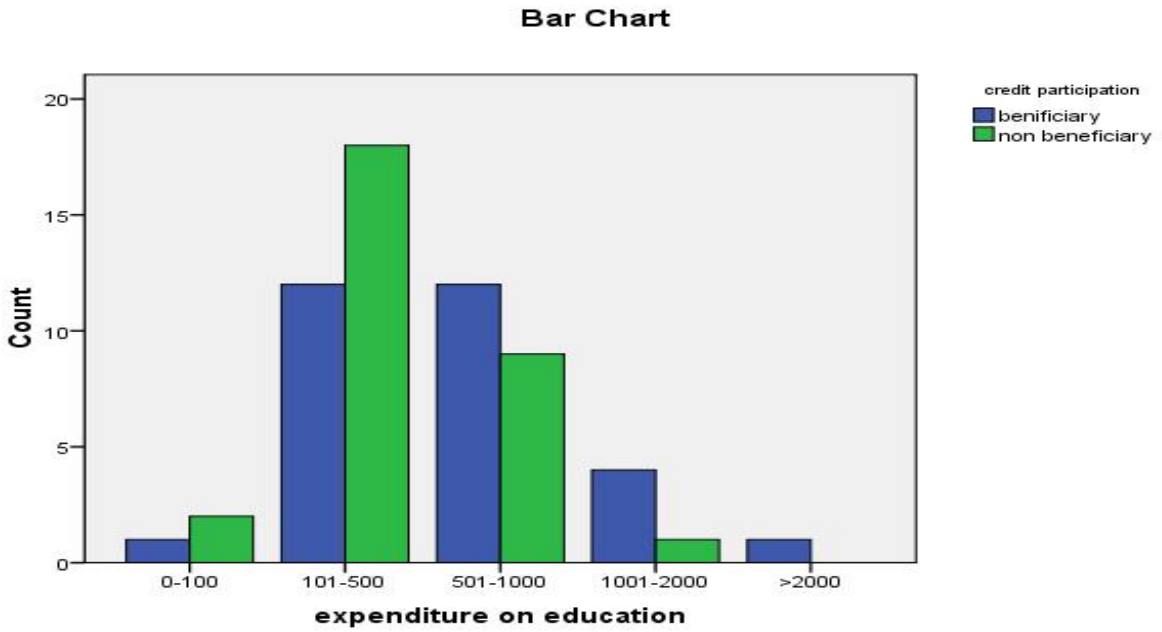
As indicate in the Table 4.6, the mean difference of children at school age between beneficiaries and non-beneficiaries is -0.567. The negative sign implies beneficiaries have less number of children at school age. The p-value indicates significant variation between beneficiaries and non-beneficiaries in terms of the number of children at school age at 10% level of significance. Non- beneficiaries have high number of children at school age as compared to beneficiaries. However, half of them are attending formal school. On other hand, 75% of beneficiaries' children those whose age is reached for child education is attending formal school. This indicates the role of microfinance credit in enabling the beneficiaries in educating their children at school age on time by raising their income and social status. So, microfinance institutions play the great role in addressing primary education for poor part of the community in rural areas.

Even though there is difference between beneficiaries and non-beneficiaries in terms of child education, the number of children attending formal school is increasing in generally due to establishment of new schools in the area; improvement of the household income and enrollment of these children newly joined school age

4.6.2. Expenditure on Education

Increased income enable the clients not only sending their children to school, it also increases school expenditure. Based on this, an attempt has been made to identify whether there are differences or not between the beneficiaries and non-beneficiaries with regard to current education expenditure per child. The educational expenditures include clothing (school uniform), exercise books and related stationeries, books; school fees or registration fees, etc. The following figure portrays expenditure on school between beneficiaries and non-beneficiaries.

Figure 4.4, Mean Annual Expenditure on Child Education.



Source: Primary Data

As indicated in the above figure, most of non-beneficiaries expend 500 birr per year on child education. This indicates that, either they don't send their children to school or the children do not get the necessary facilities for their education. On the other hand, the mean annual expenditure of beneficiaries on child education is from 500 birr up to 1000 birr. Which is high as compared to what non-beneficiaries expend; this implies that beneficiaries get higher income from the business they run using the credit received from AMFI. So, that they expend high than non-beneficiaries.

4.7. Logistic Regression

Table 4.5. Factors determining participation in microfinance credit program

Variables in the Equation		B	S.E.	Wald	Sig.	Exp(B)
	Types of asset			7.156	0.028	
	land(1)	-8.877	4.801	3.418	0.064	0.000
	livestock (2)	-11.543	4.539	6.466	0.011	0.000
	Sex(1)	-3.211	2.064	2.420	0.120	0.040
	Age	0.064	0.108	0.347	0.556	1.066
	Mstat(1)	-1.818	0.918	3.922	0.048	0.162
	Religion(1)	-.804	9.952	0.007	0.936	0.448
	Religion(2)	-1.226	9.982	0.015	0.902	0.294
	Famsiz	1.943	1.031	3.553	0.059	6.979
	Edulevel	0.275	0.384	0.513	0.074	1.317
	Depratio	-.759	0.822	0.853	0.356	0.468
	Constant	2.836	11.665	0.059	0.808	17.054

***, statistically significant at 5% and 10%*

The table summarizes the effect of variables on household engagement in microcredit function of the microfinance in the models.

From the table, asset ownership, marital status, educational level and family size are significant in the model. The likelihood ratio decreases by -8.877 as it goes from household who has land to who has livestock; similarly it further decreases for household who has different assets like livestock, land and others. In both cases the odds ratio is the same, while there is different significance levels; this could be due to the difference in unit of measurements of these asset categories are measured. This indicates household who has different types of assets (both physical and social assets) is less likely to participate in the credit program. This might be due to the primary focus of microfinance institutions is addressing the socio-economic problems of the poor part of the community not the rich part. Moreover, marital status also has significant effect on credit participation; in the above table, mstat (1) the reference category (1) shows ingle marital status. The negative B (coefficient of independent variable) shows from single to the next marital status the probability of credit participation decreases; while the probability of the next marital status (married household0 in credit participation increases by 16.2%. This indicates married households are socially acceptable; so that they have good relation with their group members. Furthermore, as family size increases by one unit, the likelihood to participate or the odds of participation in credit service increases 7 times. This is may be because of households with high number of family needs high income to meet the needs of the family. So, as a means of increasing their income, they participate in the credit program given by AMFI. Similarly, as education increases by one unit the odds of participation increases 1.137 times.

On the other hand, the effect of variables such as sex, age, religion, and dependency ratio with credit participation is not as such strong; so, they do not significantly affect credit participation.

5. CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

Access to microfinance credit services improve and diversify the income of beneficiary households. The comparisons made between the incomes of beneficiary and non-beneficiary households show that there is a significant difference between them. The study in the two branch's, Tebassie and Atsezereyaekob, demonstrate that microfinance credit service has enabled households to educate their children, afford the food needs of their family and enable them to get medical service. This also indicates that, improved incomes will encourage these households to invest in the education of their children. This is clearly seen by the fact that, the expenditure on education has been significantly high among the beneficiaries. Furthermore, the assessment made on the income generating activities the two categories demonstrate that, the beneficiary households engaged in non-farm activities in addition to farming, especially in the small scale trading sector which include: small shopping and purchasing grains during the harvesting time and sell during the cropping season.

Lack of good transport, small size of the loan, short or fixed repayment period and seasonal production are the major challenges that faced microfinance beneficiaries in their credit participation program. Lack of microfinance services could be a constraining factor to income diversification, which can be an important risk management strategy for rural households. Furthermore, microfinance services are used to spread risks when catastrophe situations occur. Such circumstances may not affect the diverse income generation activities of households equally and severely. Generally, from the findings of this study one understands that access to microfinance services can result in reduced vulnerability and improved the livelihood of the beneficiaries.

.

5.2. Recommendations

Based on the findings obtained from result and discussion, the following recommendations are made:

- The current small size loans should be increased to enable the beneficiaries engaging in large business activities.
- The current coverage of beneficiary households should be increased to reach as many poor households as possible by giving awareness about the importance of microfinance credit in improving livelihood.
- Microfinance service delivery should be flexible in terms of repayment periods and frequency of repayment in such a way that during catastrophe circumstances (drought, conflict, etc) repayments should be postponed to good season
- The group guarantee which, AMFI allows should include individuals who have no asset; since the main purpose of microfinance scheme is supporting the poor part of the community.
- Different income generation activities should be designed for different areas based on their local contexts. For instance, fodder production, fruits and vegetables production, shoat fattening and trading, dairy milk, and other petty trading activities.
- Finally, participatory research is required to gain adequate understanding of the problems faced in microfinance credit participation by setting appropriate risk management strategies and indigenous credit systems to further complement the microfinance services.

6 REFERRENCES

Aryeetey, E. (2008), “From Informal Financing to Formal Financing in Sub Saharan Africa”, lesson from linkage efforts.

Asiama, J.P. and Osei, V. (2007), “Micro finance in Ghana: An Overview“, Research Department Working Paper Bank of Ghana 07/01.

Assefa A. (2004), New Pro-poor Policies and Budgeting in the Ethiopian Economic Sector.

Charles K. (2003/Hdi0/1911 U), “The Contribution of Microfinance Institutions (Mfis) to the Economic Activities of the Youth and Women in Luwero District in Uganda”

CSA (2011), Statistical Abstract; Government of the Federal Democratic Republic of Ethiopia: Addis Ababa, Ethiopia.

Daba M. (2004), the Impact of Microfinance on Poverty Reduction: A Case Study of DebreberhanCredit and Saving Share Company.

Dejene, A., 1999. The Village Economy and Indigenous Financial Institutions

Ditcher, T. W. (2006), "Hype and Hope”: The Worrisome State of the Microcredit. http://conservation finance.wordpress.com (Accessed on 30/04/11

Gebrehiwot A. (2002), Microfinance Institutions in Ethiopia: Issues of Portfolio Risk

Gobezie, G., 2005. Regulating Microfinance in Ethiopia: Making it more Effective.

Grameen Bank (2000), [Online] Available at: www.grameen-info.org. (Accessed on16/4/2011)

Http://en.wikipedia.org/wiki/Agaro

Hulme D. and Mosley P. (1996), Finance against Poverty: Effective Institutions for Lending to Small Farmers and Micro-enterprises in Developing Countries.

Mayoux, Linda (1997). *The Magic Ingredient, Micro finance and Women’s Empowerment*: Washington, Feb. 1997.

Osmani L. and N. Khan (1998), Impact of Credit Program on the Relative Well-being.

Otero M. (1999), “The Evolution of Nongovernmental Organizations toward Financial Intermediation.”

Schreiner M and Murdoch, J. (2001), “Replicating Microfinance in the United States: Opportunities and Challenges.”

Scoones, Ian. (1998), Sustainable Rural Livelihood: Framework for Analysis.

Shimelles T&K.M. Zahidul (2009); ”Rural Financial Services and Effects of Microfinance on Agricultural Productivity and on Poverty” in Helsinki

Simanowitz, P. & Brody, S. (2004), “Realizing the potential of microfinance, id21 insights”, December, Issue -51

Sinha, S. (1998), “Micro-Credit: Impact, Targeting and Sustainability”, IDS bulletin, Vol. 29, No.4.

Tebeje&Chokole. (2007), Role and Challenges of Micro finance Institutions in Ethiopia.

Tsehay T. &Mengistu B. (2002), The Impact of Micro finance Service among poor.

WoldayAmha (2006), “Managing the growth of micro finance institutions to Reach women in Ethiopia”; Occasional paper No. 6, AEMFI, Addis Ababa

7 APPENDEX

Interview Schedule

Debre Berhan University of Agricultural economics, Interview Schedule for The Role of Microfinance Institution in Improving Rural Household income; in Case of AMFI in Tebassie District.

The purpose of this interview schedule is to gather information to write a research paper on the role of AMFI in improving the rural income of household in Tebassie District. Personal responses of interviewees would be kept confidential, and there would not be any link between status in the program me and responses.

GENERAL INFORMATION AND ADMINISTRATIVE

Date of interview ____/_____/_____

Household ID: _____ Phone No: _____

Regional State _____Administrative zone: _____

District _____ Branch _____

SECTION I: Household Characteristics, Demographic, and Socio-economic Data

1. Category of the Household [1] participant [2] Non- participant
2. How many household members do you have including you? _____
3. Please could you list all of your household members?

S/N	household member	Sex 1.Male 2.Female	Age [year]	Relationship to the head
1				
2				
3				
4				
5				

4. Household head Religion: [1] Muslim [2] Christians
5. Marital status of household head: [1] Single [2] Married [3] Widowed [4] Divorced
6. Education level of household:_____
   - ✓ Note that you have to write 0 for not educated households and year of schooling.
7. Do you have your own land? [1] Yes [2] No
8. If your response to Question No. 7 is **yes** could you please mention in detail?

No	Purpose of use	Area in hectare

1	Crop land	
3	Grazing land	
4	Coffee and forest land	
5	Fallow land	
6	Homestead	
7	Other land (specify)	
8	Total land size you have	

9. Do you have livestock? [1] Yes [2] No
10. If your response to Question No.9 is **yes**, please indicate current number of livestock you have in the following table?

Livestock Holding			Type of Livestock									
	Oxen	Cow	Bull	Calf	Heifer	Sheep	Goat	Donkey	Horse	Mule	Hens	**Total**
In number/In Unit												
Number of sold												
Price /Unit												

11. Did you produce crop in last cropping year 2012? [1] Yes [2] No
12. If your answer for Question No 11 is yes please fill the following table accordingly

Amount of the crop produced in last production year?	Type of crop										
	Carrot	Maize	Sorghum	Teff	Wheat	Barley	Millet	Rice	Vegetable	Fruit	If others specify
Total production you gained in past 12 month in quintal											
Amount sold											
Price/quintal											

13. Did you engage in off/non-farm activities? [1] Yes [2] No

14. If your answer for question number 13 is yes what types off farm/non-farm activities it? (Multiple answers are possible)

[1] Petty trade [2] Handcrafts [4] Laborer

[5] Employed [6] Fishing [8] other.....

Types of activities	Annually income
Petty trade	
Handcrafts	
Employed	
Fishing	
If others	

15. Distance from home of household to institutional infrastructure (in hour and converted to Km)

No	**Type of infrastructure**	**Distance in Km**
1	Nearest debreberhan Akufada Microfinance institution	
2	Nearest market	

16. Have you member ship of cooperative? [1] Yes [2] No
17. If your answer to question No.31 is yes, what type of cooperative and what benefit you have gained from it? State in the following table (Multiple answers are possible)

No	**Type of cooperative**	**Benefit you have gained from it**
1	Producer	
2	Marketing	
3	Consumer	
4	Worker	
5	Multi-stakeholder	
6	Non-profit community service	
7	Financial	
+	If other	

1.7. Household (family) size_____

PART TWO: ACCESS TO LOAN, LOAN UTILIZATION AND REPAYMENT

2.1 Did you have an access to credit before joining credit program?

1. Yes 2. No

2.2 If your answer for question 2.1 is yes, state the sources of the loan, amount of the loan and purpose of borrowing.

Source	Amount of loan	Purpose of borrowing

2.3 Have you repaid the loan? a. Yes b. No c. Partially

2.4 If you haven't repaid the loan, state your reasons for not repaying

_____

2.6 What guarantee did you provide for AMFI to take the loan?

a. Group responsibility
b. Guarantee of salaried individuals
c. Guarantee of individuals having assets like home, car etc.
d. Other (specify) _____

2.7 Have you and your group been willing to accept people without any assets and business activities as group members?

i. Yes ii. No

2.8 If your answer for Q. 2.7 is no, state why you have not been willing to accept people without assets as group numbers

2.9 For what purposes did you receive loan from AMFI Microfinance Institution?

Intended purpose	Amount of loan

2.10 Did you spend the total amount you borrowed on the purposes intended?

1. Yes 2. No

2.11 If your answer for the question above is no, state the other purposes on which you spent the money

_____

2.12 How do you see the interest you pay on loans? a. High b. Normal c . Low

2.13 Was the loan sufficient to carry out your purpose? 1. Yes 2. No

2.14. If your answer is no, what alternative options did you take?

_____

2.15 Has the loan repayment been suitable for you? 1. Yes 2. No

2.16 If no, state why it has not been suitable for you

Reason _____

2.17 Would you state the amount of loan you did not pay so far? _____

2.18 Have you ever been faced with problem for not repaying loans?

1. Yes 2. No

2.19 If yes, state the problems _____

PART THREE: INCOME, ASSETS AND EXPENDITURE

3.1. What is your source(s) of income? What do you do for living?

Source of Income	Income Generating Activities

3.5. Do you have a saving account? 1. Yes 2. No

3.6. If yes, how much is your mean monthly saving? _____

3.7. What is your source of money for saving?

a. Business financed by the loan
b. Another business different from that is financed by the loan
c. Gift from relatives/friends
d. Other (specify) _____